

Lock Service over Complete M-Way Tree

#topics/tree

Given a *Complete M - Way Tree*. (or M- ary Tree). Implement the following operations on it :

1. Lock(X, User ID) : Place a lock on the entire subtree, rooted at node X, by the given User ID. This means that all the nodes that are in the subtree of this node, cannot be locked by anyone,. This operation will also fail if X is in the subtree of an already locked node. This will fail if the subtree of the node X, contains a locked Node. This function returns true if successful, false otherwise
2. Unlock(X, User ID) : Releases the lock placed on Node X. A Node can only be unlocked by the same user who placed the lock.
3. Upgrade(X, User ID) : For Node X, which is unlocked, but all its descendants are locked by the same user, then that user can upgrade the lock by Locking node X. This operation should maintain the consistency of Lock and Unlock operations as well.

Constraints : $2 \leq M \leq 30$. Number of Operations $\leq 1e5$. Number of Nodes in Tree $\leq 1e5$